MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, LOW-POWER, TYPE 2N2857
JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN, silicon, low-power, ultra-high frequency transistors. Four levels of product assurance are provided for the device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1.

1.3 Maximum ratings.

P_T 1/ T_A = 25°C	P_T 2/ T_C = 25°C	V_{CBO}	V_{CEO}	V_{EBO}	I_C	T_J and T_{STG}
mW	mW	V dc	V dc	V dc	mA dc	°C
200	300	30	15	3	40	-65 to +200

1/ Derate at 1.14 mW/°C above 25°C.

2/ Derate at 1.71 mW/°C above 25°C.

1.4 Primary electrical characteristics.

	h_{FE1}	$\|h_{fe}\|$	C_{cb}	F	G_{pe}	$r_b'C_C$
Limits	V_{CE} = 1 V dc I_C = 3 mA dc	V_{CE} = 6 V dc I_C = 5 mA dc f = 100 MHz	V_{CB} = 10 V dc I_E = 0 100 kHz ≤ f ≤ 1 MHz	V_{CE} = 6 V dc I_C = 1.5 mA dc f = 450 MHz Rg = 50 ohms (see figure 3)	V_{CE} = 6 V dc I_C = 1.5 mA dc f = 450 MHz	V_{CB} = 6 V dc I_E = 2 mA dc f = 31.9 MHz (see figure 4)
			pF	dB	dB	ps
Min	30	10	---	---	12.5	4
Max	150	21	1.0	4.5	21	15

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center, ATTN: RBE-2, Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500, and as follows:

C_{cb} - Collector to base feedback capacitance.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500 and figure 1 herein.

3.3.1 Lead material and finish. Lead material shall be Kovar or Alloy 52. Lead finish shall be tinned, gold-plated, or solder-dipped. Where a choice of lead material or finish is desired, it shall be specified in the contract or purchase order (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-S-19500. At the option of the manufacturer, the following marking may be omitted from the body of the transistor:

a. Country of origin.
b. Manufacturer's identification.

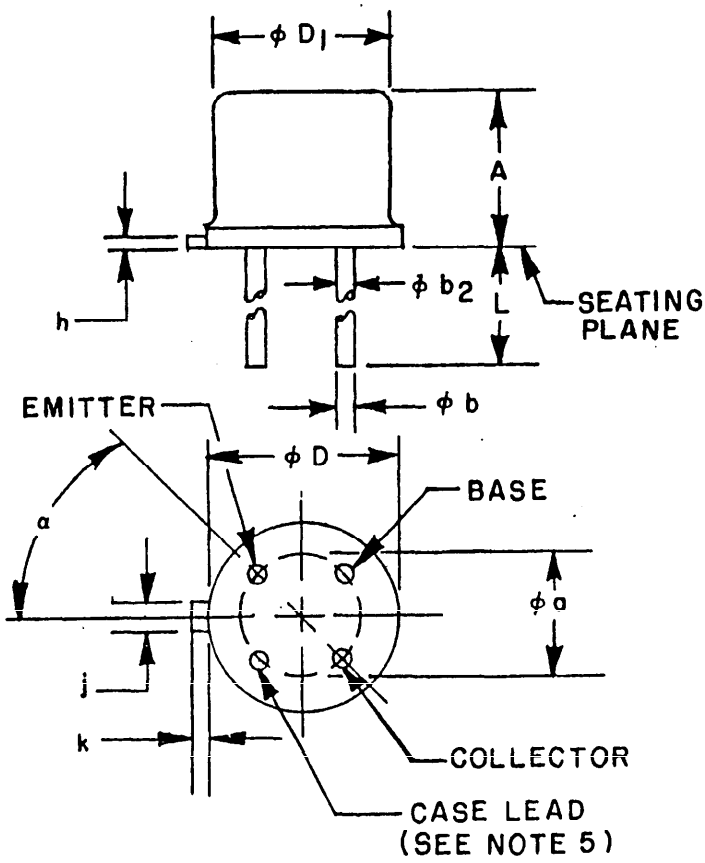
4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.3 Screening (JANS, JANTXV, and JANTX levels only). Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

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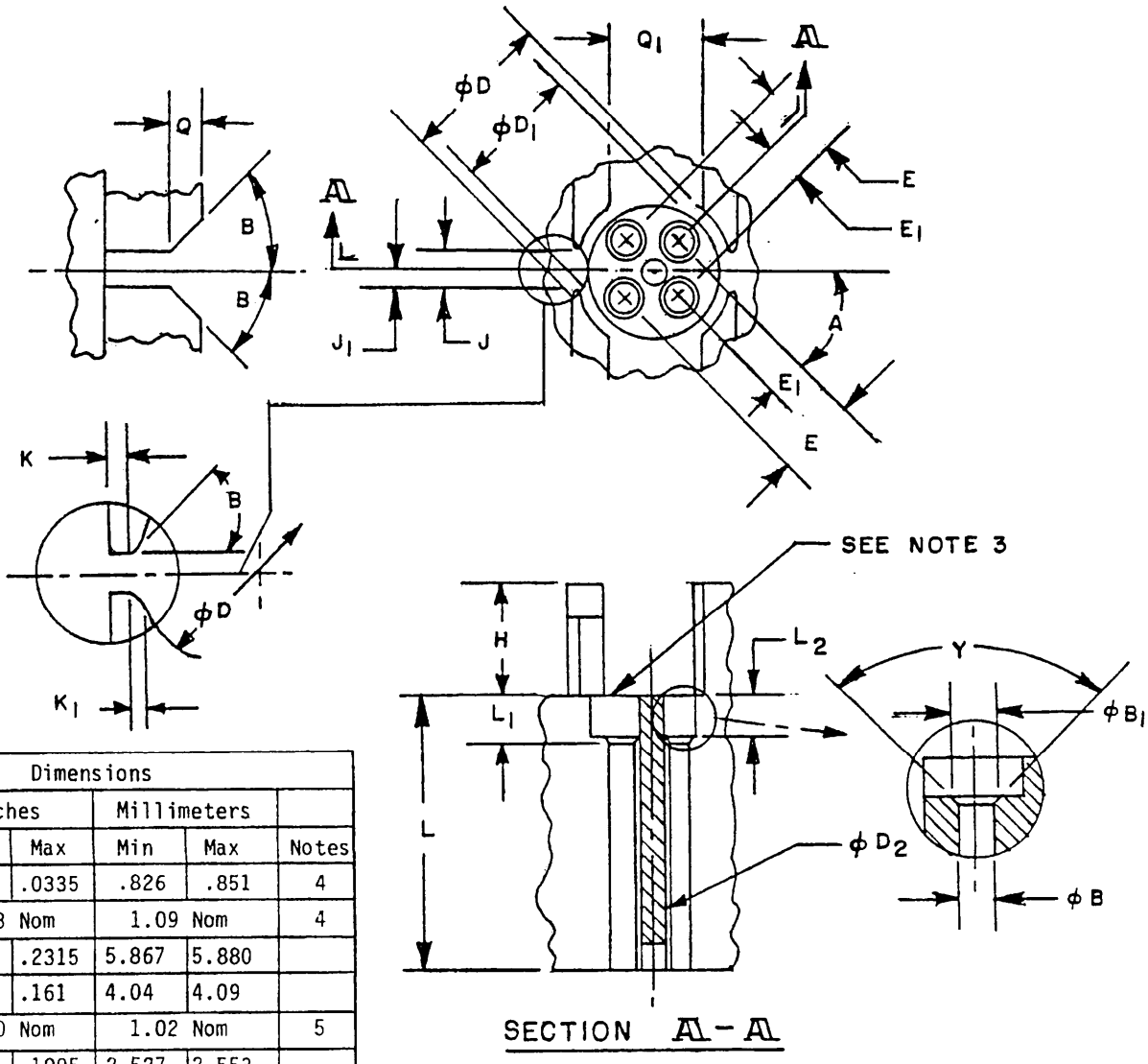


Dimensions					
Ltr	Inches		Millimeters		Notes
	Min	Max	Min	Max	
A	.170	.210	4.32	5.33	
Øa	.100 Nom		2.54 Nom		4
Øb	.016	.021	0.41	0.53	2,7
$Øb_2$	.016	.019	0.41	0.48	3,7
ØD	.209	.230	5.31	5.84	
$ØD_1$	.178	.195	4.52	4.95	
h	---	.020	---	0.51	
j	.036	.046	0.91	1.17	
k	.028	.048	0.71	1.22	6
L	.500	.750	12.70	19.05	7
α		45° Nom			

NOTES:
1. Metric equivalents are given for general information only.
2. Measured in the zone beyond .250 (6.35 mm) from the seating plane.
3. Measured in the zone .050 (1.27 mm) and .250 (6.35 mm) from the seating plane.
4. When measured in gauging plane .054 +.001, -0.000 (1.37 +0.03, -0.00 mm) below the seating plane of the transistor, maximum diameter leads shall be within .007 (0.18 mm) of their true location relative to a maximum width tab. Smaller diameter leads shall fall within the outline of the maximum diameter lead tolerance. Figure 2 preferred measured method.
5. Emitter, base, and collector leads electrically insulated from the case. The case lead is electrically connected to the case.
6. Measured from the maximum diameter of the actual device.
7. All 4 leads.

FIGURE 1. Physical dimensions.

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Dimensions					
	Inches		Millimeters		
Ltr	Min	Max	Min	Max	Notes
ØB	.0325	.0335	.826	.851	4
$ØB_1$	.043 Nom		1.09 Nom		4
ØD	.2310	.2315	5.867	5.880	
$ØD_1$	.159	.161	4.04	4.09	
$ØD_2$	.040 Nom		1.02 Nom		5
E	.0995	.1005	2.527	2.553	
E_1	.0495	.0505	1.257	1.283	
H	.145	.155	3.68	3.94	
J	.0470	.0475	1.194	1.206	
J_1	.0235	.0245	.597	.622	
K	.009	.011	.229	.279	
K_1	.005 Nom		.13 Nom		
L	.372	.378	9.45	9.60	
L_1	.054	.055	1.37	1.40	
L_2	.043 Nom		1.09 Nom		
Q	.040 Nom		1.02 Nom		
Q_1	.123	.127	3.12	3.23	
A	44.90° Min		45.10° Max		
B		45° Nom			
Y		90° Nom			

NOTES:
1. Metric equivalents are given for general information only.
2. The following gauging procedure shall be used: The device being measured shall be inserted until its seating plane is. 125 (3.18 mm) +.010 (0.25 mm) from the seating surface of the gauge. A force of 8 ±.5 oz. shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed) the seating plane of the device shall be seated against the gauge. The use of a pin straightener prior to insertion in the gauge is permissible. A spacer may be used to obtain the .125 (3.18 mm) distance from the gauge seat prior to force application.
3. These surfaces to be parallel and in same plane within ±.001 (0.03 mm).
4. Four holes.
5. Pressed in.

FIGURE 2. Gauge for lead and tab location.

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Screen (see table II of MIL-STD-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
9	I_{CES} and h_{FE1}	Not applicable
11	I_{CES}; h_{FE1}; ΔI_{CES} = 100% of initial value or 5 nA dc, whichever is greater; Δh_{FE1} = 15%	I_{CES}; h_{FE1}
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; ΔI_{CES} = 100% of initial value or 5 nA dc, whichever is greater; Δh_{FE1} = 15%	Subgroup 2 of table I herein; ΔI_{CES} = 100% of initial value or 10 nA dc, whichever is greater; Δh_{FE1} = 15%

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

V_{CB} = 15 V dc, P_T = 200 mW at T_A = 25°C ±3°C.
Note: No heat sink or forced air cooling on the devices shall be permitted.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JAN, JANTX, and JANTXV) of MIL-S-19500, and tables IIa and IIb herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and table III herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Collector to base feedback capacitance. This test shall be conducted in accordance with method 3236 of MIL-STD-750 except that the case and emitter leads shall be connected to the guard terminal of the measuring equipment.

4.5.2 Noise figure. The noise figure shall be measured using commercially available test equipment and its associated standard test procedures (see figure 3).

4.5.3 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with test method 3131 of MIL-STD-750. The following details shall apply:

a. Collector current magnitude shall be 10 mA dc.

b. Collector to emitter voltage magnitude shall be 10 V dc.

c. Reference point temperature shall be 25°C $\leq T_R \leq$ 40°C. The chosen reference temperature shall be recorded before the test is started.

d. Maximum $R_{\theta JA}$ limit shall be 0.875°C/mW.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Ordering data. Acquisition documents may specify the lead material and finish (see 3.3.1).

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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TABLE I. Group A inspection.

Inspection	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Conditions	JANS	JAN, JANTX, JANTXV		Min	Max	
Subgroup 1								
Visual and mechanical inspection	2071				---	---	---	---
Subgroup 2								
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 3 mA dc			$V_{(BR)CEO}$	15	---	V dc
Breakdown voltage, collector to base	3001	Bias condition D; I_C = 1 µA dc			$V_{(BR)CBO}$	30	---	V dc
Breakdown voltage, emitter to base	3026	Bias condition D; I_E = 10 µA dc			$V_{(BR)EBO}$	3	---	V dc
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 15 V dc			I_{CBO1}	---	10	nA dc
Forward-current transfer ratio	3076	V_{CE} = 1 V dc; I_C = 3 mA dc			h_{FE1}	30	150	---
Collector to emitter voltage (saturated)	3071	I_C = 10 mA dc; I_B = 1 mA dc			$V_{CE(sat)}$	---	0.4	V dc
Base to emitter voltage (saturated)	3066	Test condition A; I_B = 1 mA dc; I_C = 10 mA dc			$V_{BE(sat)}$	---	1.0	V dc
Collector to emitter cutoff current	3041	Bias condition C; V_{CE} = 16 V dc			I_{CES}	---	100	nA dc
Subgroup 3								
High-temperature operation:		T_A = 150°C						
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 15 V dc			I_{CBO2}	---	1.0	µA dc
Low-temperature operation:		T_A = -55°C						
Forward-current transfer ratio	3076	V_{CE} = 1 V dc; I_C = 3 mA dc			h_{FE2}	10	---	---
Subgroup 4								
Noise figure	---	V_{CE} = 6 V dc; I_C = 1.5 mA dc; f = 450 MHz; Rg = 50 ohms (case lead grounded) (see 4.5.2 and figure 3)			F	---	4.5	dB

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Conditions	JANS	JAN, JANTX, JANTXV		Min	Max	
Magnitude of common-emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 6 V dc; I_C = 5 mA dc; f = 100 MHz (case lead grounded)			$\lvert h_{fe} \rvert$	10	21	---
Small-signal short-circuit forward-current transfer ratio	3206	V_{CE} = 6 V dc; I_C = 2 mA dc (case lead floating)			h_{fe}	50	220	---
Collector to base feedback capacitance	---	V_{CB} = 10 V dc; I_E = 0; 100 kHz $\leq$ f $\leq$ 1 MHz (see 4.5.1)			C_{cb}	---	1.0	pF
Small-signal power gain	3256	V_{CE} = 6 V dc; I_C = 1.5 mA dc; f = 450 MHz			G_{pe}	12.5	21	dB
Collector-base time constant	---	V_{CB} = 6 V dc; I_E = 2 mA dc; f = 31.9 MHz (see figure 4)			$r_b'C_c$	4	15	ps
Subgroups 5, 6, and 7								
Not applicable								

1/ For LTPD numbers, see MIL-S-19500.

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TABLE IIa. Group B inspection for JANS devices.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection LTPD 1/	Small lot quality conformance inspection n/c 1/
	Method	Conditions		
Subgroup 1				
Physical dimensions	2066	See figure 1		
Subgroup 2				
Solderability	2026			
Resistance to solvents	1022			
Subgroup 3				
Thermal shock (temperature cycling)	1051			
Hermetic seal	1071			
a. Fine leak				
b. Gross leak				
Electrical measurements		See table IV, steps 1, 3, 4, and 5		
Decap internal visual design verification	2075			
Bond strength	2037			
Die shear	2017			
Subgroup 4				
Intermittent operation life	1037	V_{CB} = 10 V dc; P_T = 200 mW at T_A = 25°C ±3°C; t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles. No heat sink or forced air cooling on the devices shall be permitted.		
Electrical measurements		See table IV, steps 1, 3, 4, 5, and 8		
Subgroup 5				
Accelerated steady state operating life	1027	V_{CB} = 10 V dc; T_A = 125°C ±25°C for 96 hours. P_T = 200 mW at 100°C or adjusted as required by the chosen T_A to give an average lot T_J = 275°C. Marking legibility requirements shall not apply.		

See footnote at end of table.

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TABLE IIa. Group B inspection for JANS devices - Continued.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection LTPD 1/	Small lot quality conformance inspection n/c 1/
	Method	Conditions		
Electrical measurements		See table IV, steps 1, 3, 4, 5, 6, 7, and 8		
Bond strength (Al-Au interconnects only)	2037			
Subgroup 6				
Not applicable				

1/ For LTPD numbers, see MIL-S-19500.

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TABLE IIb. Group B inspection for JAN, JANTX, and JANTXV devices.

Inspection	MIL-STD-750		LTPD 1/
	Method	Conditions	
Subgroup 1			
Solderability	2026		
Resistance to solvents	1022		
Subgroup 2			
Thermal shock (temperature cycling)	1051		
Hermetic seal	1071		
a. Fine leak			
b. Gross leak			
Electrical measurements		See table IV, steps 1, 3, and 5	
Subgroup 3			
Steady-state operation life	1027	V_{CB} = 15 V dc; P_T = 200 mW at T_A = 25°C ±3°C. No heat sink or forced air cooling on the devices shall be permitted.	
Electrical measurements		See table IV, steps 2, 5, and 7	
Subgroup 4			
Decap internal visual design verification	2075		
Bond strength	2037	Test condition A.	
Subgroup 5			
Not applicable			
Subgroup 6			
High-temperature life (nonoperating)	1032	T_A = 200°C	
Electrical measurements		See table IV, steps 2, 5, and 7	

1/ For LTPD numbers, see MIL-S-19500.

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TABLE III. Group C inspection (all quality levels).

Inspection	MIL-STD-750		LTPD 1/
	Method	Conditions	
Subgroup 1			
Physical dimensions	2066	See figure 1	
Subgroup 2			
Thermal shock (glass strain)	1056	Test condition A	
Terminal strength	2036	Test condition E	
Hermetic seal	1071		
a. Fine leak			
b. Gross leak			
Moisture resistance	1021		
External visual	2071		
Electrical measurements		See table IV, steps 1, 3, 4, and 5 (JANS) and steps 1, 3, and 5 (JAN, JANTX, and JANTXV)	
Subgroup 3			
Shock	2016		
Vibration, variable frequency	2056		
Constant acceleration	2006		
Electrical measurements		See table IV, steps 1, 3, 4, and 5 (JANS) and steps 1, 3, and 5 (JAN, JANTX, and JANTXV)	
Subgroup 4			
Salt atmosphere (corrosion)	1041		
Subgroup 5			
Not applicable			
Subgroup 6			
Steady-state operation life	1026	V_{CB} = 15 V dc; P_T = 200 mW at T_A = 25°C ±3°C. No heat sink or forced air cooling on the devices shall be permitted.	
Electrical measurements		See table IV, steps 1, 3, 4, 5, 6, and 7 (JANS) and steps 2, 5, and 7 (JAN, JANTX, and JANTXV)	
Subgroup 7			
Thermal resistance	3131	See 4.5.3	

1/ For LTPD numbers, see MIL-S-19500.

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TABLE IV. Group B and C electrical measurements.

Step	Inspection	MIL-STD-750 Method	MIL-STD-750 Conditions	Symbol	Limits Min	Limits Max	Unit
1.	Collector to base cutoff current	3036	Bias condition D; V_{CB} = 15 V dc	I_{CBO1}	---	10	nA dc
2.	Collector to base cutoff current	3036	Bias condition D; V_{CB} = 15 V dc	I_{CBO1}	---	20	nA dc
3.	Collector to emitter voltage (saturated)	3071	I_C = 10 mA dc; I_B = 1.0 mA dc	$V_{CE(sat)}$	---	0.4	V dc
4.	Base emitter voltage (saturated)	3066	Test condition A; I_B = 1 mA dc; I_C = 10 mA dc	$V_{BE(sat)}$	---	1.0	V dc
5.	Forward-current transfer ratio	3076	V_{CE} = 1 V dc; I_C = 3 mA dc	h_{FE1}	30	150	---
6.	Collector to base cutoff current	3036	Bias condition D; V_{CB} = 15 V dc	ΔI_{CBO1}	100% of initial value or 5 nA dc, whichever is greater.		
7.	Forward-current transfer ratio	3076	V_{CE} = 1.0 V dc; I_C = 3 mA dc	Δh_{FE1}	±25% change from initial value.		
8.	Collector to emitter voltage (saturated)	3071	I_C = 10 mA dc; I_B = 1.0 mA dc	$\Delta V_{CE(sat)}$	±50 mV dc change from initial value.		

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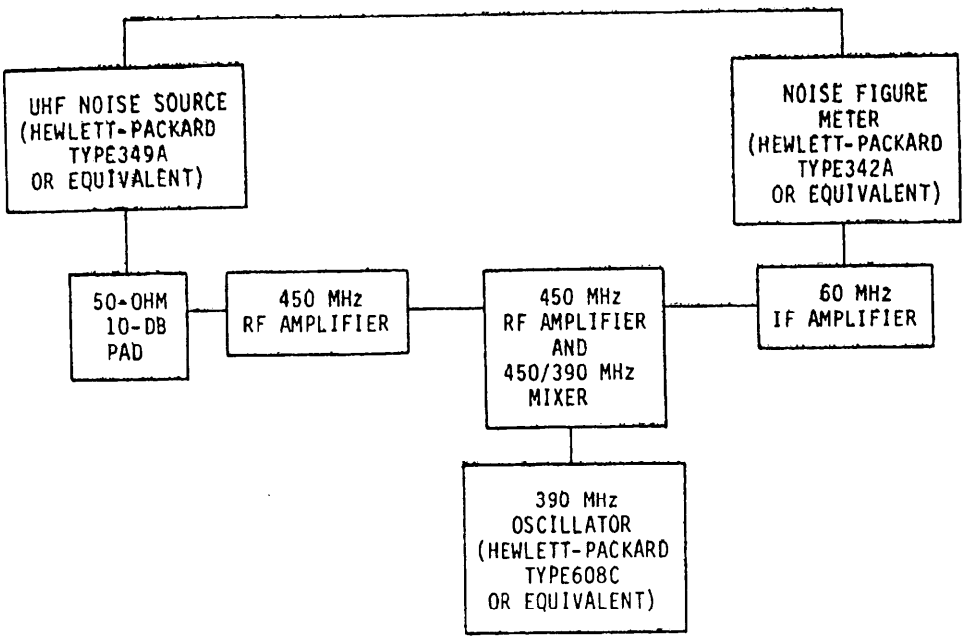
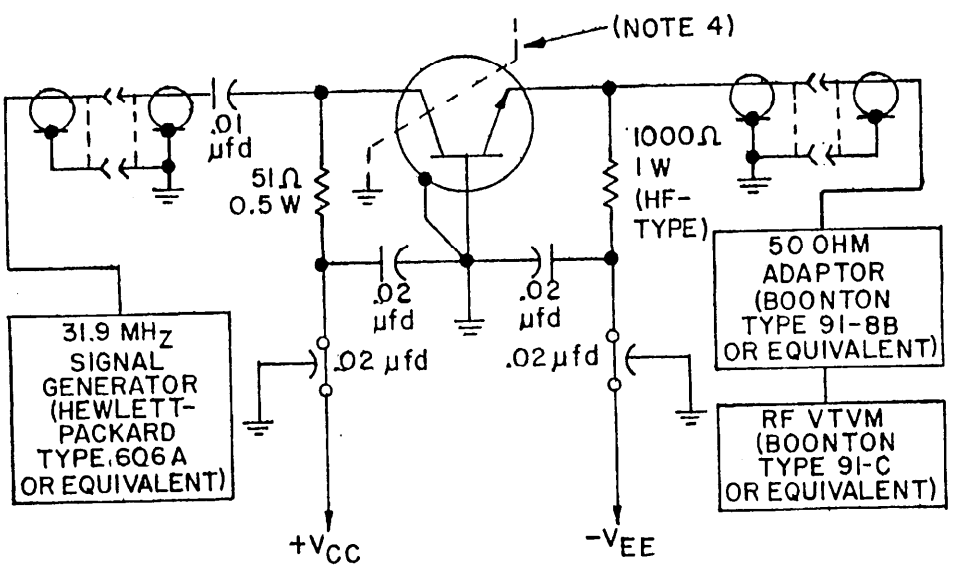


FIGURE 3. Block diagram for noise-figure test.



Procedure:

1. With a short circuit applied between the collector and emitter terminals adjust 31.9 MHz input for 0.5 volt RMS at emitter terminal of transistor.
2. After removing the short circuit between the collector and emitter circuit insert unit to be tested and adjust V_{CC} and V_{EE} for V_{CB} = 6 V dc I_E = 2 mA dc.
3. Read $r_b'C_c$ on RF voltmeter scale ($r_b'C_c$ in picoseconds = 10 times meter indication in millivolts) (1 millivolt = 10 picoseconds).
4. External interlead shield to isolate the collector lead from the emitter and base leads.

FIGURE 4. Collector-base time constant test circuit (an equivalent circuit may be used).

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Army - ER
Navy - EC
Air Force - 17
NASA - MSFC-EB02

Review activities:
Army - MI, AR
Navy - SH, TD
Air Force - 11, 85
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC, OS
Air Force - 19

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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